Regulation: R20	II / IV - B.Tech. I - Semester								
CIVIL ENGINEERING									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2020-21 admitted Batch onwards)									
Course Code	**Course Name**	**Category**	**Cr**	**L**	**T**	**P**	**Int. Marks**	**Ext. Marks**	**Total Marks**
B20 BS 2101	Numerical Methods & Vector Calculus	BS	3	3	0	0	30	70	100
B20 CE 2101	Strength of Materials	PC	3	3	0	0	30	70	100
B20 CE 2102	Environmental Engineering	PC	3	3	0	0	30	70	100
B20 CE 2103	Fluid mechanics	PC	3	3	0	0	30	70	100
B20 CE 2104	Surveying	PC	3	3	0	0	30	70	100
B20 CE 2105	Strength of materials lab	PC	1.5	0	0	3	15	35	50
B20 CE 2106	Surveying field work	PC	1.5	0	0	3	15	35	50
B20 CE 2107	Computer Aided Drawing	PC	1.5	0	0	3	15	35	50
B20 CE 2108	Computer Fundamentals and MS – Office	SOC	2	1	0	2	--	50	50
B20 MC 2102	Professional Ethics and Human Values	MC	0	2	0	0	0	0	0
		TOTAL	**21.5**	**18**	**0**	**11**	**195**	**505**	**700**

Code	Category	L	T	P	C	I.M	E.M	Exam
B20BS2101	**BS**	**3**	**--**	**--**	**3**	30	**70**	**3 Hrs.**

NUMERICAL METHODS & VECTOR CALCULUS

(Common to CE, CSE, EEE & IT)

Pre-requisites: Basic concepts of calculus.

Course Objectives: Students are expected to learn

1	Numerical methods to solve algebraic and transcendental equations and the concept of interpolation and its use for equally and unequally spaced data points,
2	Methods for numerical evaluation of integrals and for solving first order ODEs.
3	Concepts of double, triple integrals and its applications.
4	Concepts of Gradient, divergence, curl.
5	Vector integral theorems.

Course Out Comes: At the end of the course students will be able to

S. No	OUT COME	Knowledge Level
1	Determine a real root of an algebraic or transcendental equation. Fit an interpolation formula and perform interpolation for equally spaced and unequally spaced data.	K3
2	Evaluate numerically certain definite integrals. Solve a first order ordinary differential equation by Euler and RK methods.	K3
3	Evaluate double integrals and determine the areas.	K3
4	Evaluate triple integrals and determine the volumes.	K3
5	Find the gradient of a scalar function, divergence and curl of a vector function.	K3
6	Solve simple problems using vector integral theorems.	K3

SYLLABUS

UNIT-I (10 Hrs)	Solution of Algebraic and Transcendental Equations: Introduction, Bisection method, Method of false position, Iteration method &Newton-Raphson method. Interpolation: Introduction, forward differences, backward differences, and Central differences. Differences of a polynomial, Newton's forward, and backward interpolation formulae. Interpolation with unequal intervals: Newton's divided difference and Lagrange interpolation formulae.
UNIT-II (10 Hrs)	Numerical Integration and solution of Ordinary Differential equations: Trapezoidal rule, Simpson's $1/3^{rd}$ rule. Solution of first order ordinary differential equations subjected to initial conditions by Taylor's method, Picard's method, Euler's method, Modified Euler's method and Fourth order Runge-Kutta method.

UNIT-III (12 Hrs)	Multiple integrals: Double and triple integrals, Change of order of integration. Change of variables, applications to finding Areas and Volumes.
UNIT-IV (10 Hrs)	Vector differentiation**:** Scalar and vector point functions, Vector Differentiation, Gradient, Directional derivative, Divergence, Curl, Scalar Potential.
UNIT-V (14 Hrs)	Vector Integration: Line integral, Work done; Area, Surface and volume integrals, Vector integral theorems: Greens, Stokes, and Gauss Divergence theorems (without proof).

TEXTBOOKS:	
1.	Scope and Treatment as in “Higher Engineering Mathematics”, by Dr.B.S.Grewal, 43rd Edition, Khanna Publishers.

REFERENCE BOOKS:	
1.	Advanced Engineering Mathematics, by Erwin Kreyszig,Wiley
2.	Higher Engineering Mathematics, by B.V.Ramana, Tata Mc Graw Hill Company.
3.	A text book of Engineering Mathematics, by N.P.Bali and Dr. Manish Goyal, Lakshmi Publications.
4.	Peter O’ Neil, Advanced Engineering Mathematics, Cengage.
5.	Advanced Engineering Mathematics, by H.K.Dass, S.ChandCompany.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CE2101	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

STRENGTH OF MATERIALS

(For CE)

Course Objectives: Students are expected to learn

1.	The concepts of stress, strain and elastic constants and their relations.
2.	Shear force, bending moment, deflections and torsion induced, shear stresses and bending stresses developed for different sections of beams and shafts
3.	Concept of principal stresses and principal strains
4.	Stresses and strains induced in columns & cylinders

Course Outcomes: At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	Summarize the behavior of basic materials under the influence of different external loading conditions and support conditions.	K3
2.	Determine shear Force and Bending moments in statically determinate Beams and draw the Diagrams.	K3
3.	Calculate the bending stresses & shear stresses in structural Members.	K3
4.	Determine the Principal Stresses & strains under different loadings and also examine the basic methods to find slope and deflection of beams subjected to different Loads.	K3
5.	Determine the crippling load for columns with different end conditions.	K3

SYLLABUS

UNIT-I (12Hrs)	Definitions of stress and strain – types of stresses and strains – Elasticity – Hooke's law – Stress – Strain diagram for Mild steel – working stress- factor of safety- Lateral strain – Poisson's ratio and volumetric strain – Elastic Moduli and the relationship between them – Bars of varying section– composite bars– temperature stresses.
UNIT-II (8 Hrs)	Shear Force &Bending Moment Diagrams: Definition of beam – Types of beams – concept of SF and BM – SF &BM diagrams for cantilever, SS and overhanging beams subjected point loads, UDL, Uniformly varying loads and combination of these loads–point of contra flexure–Relationship b/w S.F,BM and rate of loading.
UNIT-III (10 Hrs)	Flexural Stresses &Shear Stresses: *Flexural Stresses*: Theory of simple Bending– Assumptions–Derivation of Bending equationNeutralaxis–Determinationofbendingstresses–sectionmodulus of rectangular & Circular sections – Design of simple beam sections. *Shear Stresses:* Derivation of shear stress formula – shear stress distribution across various beam sections, Definition of shear centre.
UNIT-IV (10 Hrs)	Principal Stresses and Strains: Introduction-stresses on an inclined section of a bar under axial loading-compound stresses-Normal and tangential stresses on an inclined plane for biaxial stresses-Two perpendicular normal stresses accompanied by a state of simple shear-

	Mohr's circle of stress-Principal planes and principal stresses DeflectionsofBeams:(i)Cantilever(ii)Simplysupportedand(iii)Overhangingbeamsusing(a) Double integration and(b)Macaulay's method
UNIT-V (10 Hrs)	Columns & Struts: Introduction – short, medium and long columns – axially loaded compression members –crushing load–Buckling load (or) critical load (or) crippling load – Euler's theory for long columns – Assumptions – Derivations of Euler's critical load formula for variousendconditions–Effectivelengthofcolumn–slendernessratio–limitations of Euler'sTheory– Rankine formula – for both long and short columns – column subjected to Eccentric loading– Euler's Method.

Textbooks:

1.	S. Ramamrutham, Strength of materials, Dhanpatrai publishers
2.	Vazraniand Ratwani ,Strengthofmaterials,KhannaPublishers
3.	T.D.GunneswaraRao and AndalMudimby, Strength of Materials Fundamentals and Applications, Cambridge University Press

Reference Books:

1.	Timoshenko and Young, Elements of strength of materials, East West press private Ltd
2.	Popov, Introduction to mechanics of solids, Engle woodcliffsN.JPrenticeHall
3.	Dr.R.K.Bansal, Strength of materials, Laxmi Publications

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CE2102	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

ENVIRONMENTAL ENGINEERING

(For CE)

Course Objectives: Students are expected to learn

1.	Outline planning and the design of water supply systems for a community/town/city and selection of source based on quality and quantity
2.	Design of water treatment plant for a village/city
3.	Impart knowledge on design of water distribution network
4.	Design of sewers and plumbing system for buildings
5.	Design of Sewage Treatment Plant

Course Outcomes At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	Select a source based on quality and quantity and Estimate design population and water demand	K3
2.	Apply the principles of water treatment methods and design unit operations	K3
3.	Explain the collection, conveyance and distribution aspects of water	K2
4.	Explain sewerage, house plumbing, preliminary and primary treatment concepts of wastewater	K2
5.	Make use of sewage treatment methods and design secondary treatment unit operations	K3

SYLLABUS

UNIT-I (10Hrs)	Introduction: Importance and Necessity of Protected Water Supply systems, Water bornediseases,Flowchartofpublicwatersupplysystem,RoleofEnvironmentalEngineer,Evolutionof water supply system Water Demand and Quantity Estimation: Estimation of water demand for a town or city, Per capita Demand and factors influencing it - Types of water demands and its variations-factorsaffectingwaterdemand,DesignPeriod,FactorsaffectingtheDesignperiod,Populationforecasting- arithmetic, geometric, incremental increase method Sources of Water: Lakes, Rivers, Comparison of sources with reference to quality, quantity and other considerations- Ground water sources: springs, Wells and Infiltration galleries, Characteristics of water– Physical, Chemical and Biological characteristics and WHO guidelines for drinking water - IS 10500 2012.
UNIT-II (10 Hrs)	Treatment of Water: Treatment methods: Theory of Sedimentation, Coagulation, Filtration. Disinfection: Theory of disinfection-Chlorination and other Disinfection methods. Removal of color and odors- Removal of Iron and Manganese - Adsorption-Fluoridation and deflouridation–Reverse Osmosis
UNIT-III (10 Hrs)	Collection and Conveyance of Water: Factors governing the selection of the intake structure, Conveyance of Water: Gravity and Pressure conduits, Types of Pipes, Pipe

	Materials, Pipe joints, Design aspects of pipe lines, Laying and testing of pipe lines. Distribution of Water: Methods of Distribution system, Layouts of Distribution networks, Water main appurtenances - Sluice valves, Pressure relief valves, air valves, check valves, hydrants, and water meters.
UNIT-IV (10 Hrs)	Sewerage: Estimation of sewage flow and storm water drainage – fluctuations – types of sewers -design of circular sewers. Sewer appurtenances – cleaning and ventilation of sewers. Sewage pumps. House Plumbing: Systems of plumbing-sanitary fittings and other accessories– one pipe and two pipe systems Sewage characteristics –Characteristics of sewage - BOD equations. COD and BOD. Treatment of Sewage: Preliminary and Primary treatment units
UNIT-V (10 Hrs)	Treatment of Sewage: Secondary treatment: Activated Sludge Process, Principles, designs, and operational problems Trickling Filters classification – design, operation and maintenance problems. Ultimate Disposal of sewage: Methods of disposal – disposal into water bodies-Oxygen Sag Curve- Disposal into sea, disposal on land, Sewage sickness. Sludge digestion and drying.

Textbooks:	
1.	Environmental Engineering – Howard S. Peavy, Donald R. Rowe, Teorge George Tchobanoglus – Mc-Graw-Hill Book Company, New Delhi, 1985.
2.	Rural Municipal and Industrial water management, KVSG Murali Karishna, Environmental Protection Society, Kakinada, 2021.
3.	Industrial Water and Wastewater Management, K.V.S.G. Murali Krishna, Paramount Publications, Visakhapatnam, 2018.
4.	Elements of Environmental Engineering – K. N. Duggal, S. Chand & Company Ltd., New Delhi, 2012.

Reference Books:	
1.	Water Supply Engineering – P. N. Modi.
2.	Water Supply Engineering – B. C. Punmia
3.	Water Supply and Sanitary Engineering – G. S. Birdie and J. S. Birdie
4.	Environmental Engineering, Ruth F. Weiner and Robin Matthews – 4th EditionElsevier,2003
5.	Environmental Engineering, D.Srinivasan,PHILearningPrivateLimited,NewDelhi,2011

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CE2103	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

FLUID MECHANICS

(For CE)

Course Objectives: Students are expected to learn

1.	To develop an insight into engineering problems related to fluids
2.	Learn about the pressure at a point, forces on fluid element and solve complex problems in engineering
3.	Know different types of fluid flows and apply the principles of conservations of mass, momentum and energy
4.	Know the Discharge, energy losses and water hammer in flow through pipes
5.	Apply conservation principles of mass, momentum and energy on fluids using control volume approaches

Course Outcomes: At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	Determine the physical properties of fluids and different types of forces acting on a fluid element extended to forces on various gates.	K3
2.	Determine the forces that are acting on immersed bodies in static fluids through Application of buoyancy and floatation	K3
3.	Apply conservation principles of mass, momentum and energy on fluids using Control volume approaches.	K3
4.	Calculate the force exerted by the fluid on bends, nozzles using impulse momentum principle	K3
5.	Determine the shear stress, Velocity, loss of head in Laminar flow through circular Pipes and Turbulent flow for rough and smooth pipes	K3

SYLLABUS

UNIT-I (10Hrs)	Basic Fluid Properties: Definition of Fluid, basic properties of fluid, Viscosity-Newton's Law of Viscosity, Capillarity and Surface Tension. Fluid Pressure: Fluid Pressure at a point, Pascal's law, Variation of pressure with elevation, Hydrostatic law, Absolute, Gauge and Vacuum Pressures. Pressure measurement– Piezometers, Manometers and Pressure Gauges. Centre of Pressure, Forces on submerged surfaces, crest gates and lock gates.
UNIT-II (10Hrs)	Buoyancy and Floatation: Archimedes Principle- Buoyancy &Floatation-Stability of Floating Bodies-Centre of Buoyancy– Meta centric Height and its Determination. Fluid Kinematics: Types of fluid flow, Velocity, Rate of flow, Continuity Equation ,Streamline, Path line, Streak line, Local, Convective and Total Acceleration; One & TwoDimensionalFlows.StreamFunction,VelocityPotential-Rotational&IrrotationalFlows, Laplace Equation, Flow net.

UNIT-III (10Hrs)	Fluid Dynamics: Energy possessed by fluid in motion, Euler's equation of motion- Bernoulli'sequation.Energycorrectionfactor.Venturimeter, orifice meter and nozzle meter. Flow through orifices and mouth pieces: Types of orifices and mouth pieces, coefficient of contraction, velocity and discharge. Flow through notches and weirs: Types of notches and weirs, Measurement of discharge
UNIT-IV (10Hrs)	Impulse momentum equation – Momentum correction factor, Forces on pipe bends and reducers .Angular Momentum– Torque and work done; Sprinkler Problems. Laminar Flow: Relation between shear and Pressure Gradients in Laminar Flow; Reynold's experiment; Critical velocity; Steady laminar flow through a circular pipe – Hagen Poiseuille's Law. Turbulent flow: shear stress in Turbulent flow, prandtl mixing length theory, Velocity distribution in pipes, Hydro-dynamically smooth and rough boundaries, Velocity distribution in rough and smooth pipes, Resistance of smooth and rough pipes.
UNIT-V (8Hrs)	Flow through pipes: Flow measurement through pipes –head due to friction – Darcy – Weisbach equation, minor losses, Total Energy Line, Hydraulic Gradient Line. Pipes in Series, pipes in parallel. Problems on Two reservoir and three reservoir lows. Water hammer, surge Tanks.

TextBooks:

1.	Fluid Mechanics and Hydraulic Machinery by P.N. Modi &S.M. Seth, Standard Book House
2.	Fluid Mechanics by A.K.Jain, Khanna Publishers.
3.	Fluid Mechanics byV.L.Streeter,TATAMcGraw Hill

ReferenceBooks:

1.	Engineering Fluid Mechanics by K.L.Kumar, S.Chand &Co.
2.	Fluid Mechanics and Hydraulic Machines by R.K.Bansal, Laxmi Publications.
3.	FMWhite,FluidMechanics,TataMcGrawHillPublication2011
4.	Relevant NPTEL Courses.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CE2104	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

SURVEYING

(For CE)

Course Objectives: Students are expected to learn

1.	principle and methods of surveying
2.	Measure horizontal and vertical-distances and angles
3.	Perform calculations based on the observation.
4.	Identification of source of errors and rectification methods
5.	Apply surveying principles to determine areas and volumes and setting out curves
6.	Use modern surveying equipment's for accurate results

Course Outcomes At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	Measure distances and angles using instruments	K3
2.	Measure levels and plotting of levels in tables, Interpret survey data and compute areas and volumes	K3
3.	Understand the working principles of Theodolite, measurement of horizontal and vertical angles using Theodolite	K3
4.	Calculate distance between points indirectly by using Theodolite (Tacheometric Surveying) and also learns plotting of simple curves	K3
5.	Use modern surveying techniques and instruments	K3

SYLLABUS

UNIT-I (10Hrs)	Introduction and Basic Concepts**:** Introduction, Objectives, classification and principles of surveying, surveying accessories. Introduction to Compass, leveling and Plane table surveying. Measurement of Distances and Directions Linear distances-Approximate methods, Direct Methods-Chains-Tapes, ranging, Tape corrections Prismatic Compass-Bearings, included angles, Local Attraction, Magnetic Declination, and dip –W.C.B systems and Q.B system of locating bearings (simple problems only)
UNIT-II (10 Hrs)	Leveling- Types of levels, temporary and permanent adjustments, methods of leveling, booking and Determination of levels, Effect of Curvature of Earth and Refraction Contouring-Characteristics and uses of Contours, methods of contour surveying. Areas-Determination of areas consisting of irregular boundary and regular boundary. Volumes **-**Determination of volume of earth work in cutting and embankments for level section, volume of borrow pits
UNIT-III (10 Hrs)	Theodolite Surveying: Types of Theodolite, temporary adjustments, measurement of horizontal angle by repetition method and reiteration method, measurement of vertical

	Angle, Trigonometrically leveling when base is accessible and inaccessible. Traversing: Methods of traversing, traverse computations and adjustments, Introduction to Omitted measurements.(simple problems only)
UNIT-IV (8 Hrs)	Tachometric Surveying: Principles of Tachometry, stadia and tangential methods of Tachometry Curves: Types of curves and their necessity, elements of simple, compound, reverse curves
UNIT-V (12 Hrs)	Modern Surveying Methods: Total Station and Global Positioning System Basic principles, classifications, applications, comparison with conventional surveying. Electromagnetic wave theory - electromagnetic distance measuring system - principle of working and EDM instruments, Components of GPS – space segment, control segment and user segment, reference systems satellite orbits ,GPS observations. Applications of GPS, Photogrammetric surveying applications and the limitations

Textbooks:	
1.	Surveying (Vol –1, 2& 3), byB.C.Punmia, Ashok KumarJain andArunKumar Jain-Laxmi Publications(P)ltd.,NewDelhi
2.	ChandraAM,“Plane Surveying and Higher Surveying”,NewageInternationalPvt.Ltd., Publishers,NewDelhi
3.	DuggalS K, “Surveying(Vol–1&2),TataMcGrawHillPublishingCo.Ltd.NewDelhi

Reference Books:	
1.	Arthur RBenton and PhilipJ Taety, Elements of Plane Surveying, McGrawHill
2.	SurveyingandlevellingbyR.Subramanian,Oxforduniversitypress,NewDelhi
3.	AroraKR“SurveyingVol1,2&3),Standard Book House, Delhi

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CE2105	**PC**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3 Hrs.**

STRENGTH OF MATERIAL SLAB

(For CE)

Course Objectives: Student shall be able to

1.	Test steel and wood for assessing their physical properties
2.	know the quality of steel and wood
3.	know the arch action in beams

Course Outcomes: After the completion of the course student should be able to

S.No	Outcome	Knowledge Level
1.	Conduct test and find Physical properties of steel and wood	K3
2.	Design the specimens for assessing particular property of the materials with Available machines	K3
3.	Decide the range of machine and set the machine accordingly by suitable modifications	K3
4.	Design experiments making use of various techniques of load measuring or deformation measuring instruments	K3

LISTOFEXPERIMENTS

1.	Tension test on Mild Steel bar
2.	Tension test on HYSD Steel bar
3.	To determine Compressive strength of wood (parallel and perpendicular to grains)
4.	To determine Modulus of rigidity of a spring
5.	To determine Brinell's hardness &Rockwell hardness on mild steel and Aluminum Specimen
6.	To determine Vicker's hardness on mild steel and Aluminum Specimen
7.	To determine Impact strength of mild steel by Charpy test
8.	To determine Impact strength of mild steel by Izod test
9.	To determine shear strength of mild steel by Double shear test
10.	Bending test on simply supported Steel /Wooden beams
11.	Bending test on cantilever Steel/ Wooden beams
12.	Draw linear arch for the given two/three point loads
13.	Verification of Maxwell's reciprocal theorem on simply supported beam

ReferenceBooks:

1.	S.Ramamrutham,Strength of materials, Dhanpatrai publishers
2.	Dr.R.K.Bansal,Strength of materials, Laxmi Publications

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CE2106	**PC**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3 Hrs.**

SURVEYING FIELD WORK

(For CE)

Course Objectives: Student shall be able to

1.	Understand and apply the basic methods of Chain Surveying and Compass Surveying
2.	Summarize the different methods of Plane Table Surveying
3.	Observe and report the different types of Leveling
4.	Study and identify the different methods of Theodolite surveying
5.	Describe the procedures available in Total Station Instrument for surveying

Course Outcomes : After the completion of the course student should be able to

S.No	Outcome	Knowledge Level
1.	Apply the linear measurement in simple Boundary Surveys.	K3
2.	Identify direction of any line using compass survey	K3
3.	Relate the importance of Theodolite in Surveying	K3
4.	Apply Concepts of Tachometry in Surveying	K3
5.	Use the Total Station in Surveying	K3

LISTOFEXPERIMENTS

1.	Study of chains and its accessories, Aligning, Ranging and Chaining and Chain Traversing
2.	Compass Traversing– Measuring Bearings & arriving included angles
3.	Levelling: Longitudinal & Cross section levelling for an open traverse
4.	Theodolite: Distance between two in-accessible points by Theodolite
5.	Trigonometric Leveling- Heights and distance problem
6.	Determination of Heights and distance using Principles of tacheometric surveying
7.	Setting out a simple circular curve
8.	Determine area using total station
9.	Traversing using total station
10.	Contouring using total station
11.	Stakeout using total station
12.	Distance, gradient ,diff ,height between two inaccessible points using total station

ReferenceBooks:

1.	Arthur R Benton and Philip J Taety, Elements of Plane Surveying ,McGrawHill
2.	SurveyingandlevellingbyR.Subramanian,Oxforduniversitypress,NewDelhi
3.	AroraKR“SurveyingVol1,2&3),Standard Book House, Delhi

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CE2107	**PC**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3 Hrs.**

COMPUTER AIDED DRAWING

(For CE)

Course Objectives: Students are expected to learn

1.	Increase ability to communicate with the people.
2.	Learn to sketch and take field dimensions.
3.	Learn basic Auto CAD skills.
4.	Learn basic engineering drawing formats.
5.	Prepare the student for future Engineering positions.

Course Outcomes: At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	Perform basic sketching techniques will improve.	K2
2.	Use architectural and engineering scales	K2
3.	Produce engineered drawings will improve.	K2
4.	Convert sketches to engineered drawings	K2
5.	Familiar with office practice and standards.	K2
6.	Familiar with Auto CAD two dimensional drawings.	K2
7.	Develop good communication skills and team work.	K2

SYLLABUS

Fundamentals of Computers

1.	Introduction
2.	Computer Hardware and Software Concepts
3.	Introduction of Personal Computer and Operating Systems WINDOWS-XP, Windows-7,FileManagement
4.	Drawing using AutoCAD
5.	Starting a New Drawing/Opening an existing drawing
6.	Drawing Commands
7.	Hatching Command Text (multi-line & single line)and Formatting Text Styles
8.	View Commands & Drawing Settings and Aids
9.	Modify Commands
10.	Dimension Command Formatting Dimension Style and Multi-leader Style
11.	Drawing Settings and Aids
12.	Saving and Plot
13.	Simple Building Drawing

Reference Books:

1.	Learning Auto CAD2010 Volume-I, Autodesk.
2.	AutoCAD2013 fundan1entals-Elisen1oss, SDC Publications

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CE2108	**SOC**	**1**	**--**	**2**	**2**	**--**	**50**	**3 Hrs.**

COMPUTER FUNDAMENTALS AND MS – OFFICE

Course Objectives: Students are expected to learn

1.	Basics of the computer hardware and its functions
2.	Basics of the computer software programs and its functions
3.	MS-Word applications
4.	MS-Excel applications
5.	MS-Power point applications

Course Outcomes: At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	Know the fundamentals of computer hardware and software	K2
2.	Apply the MS Word for practical applications	K3
3.	Apply the MS Excel for practical applications	K3
4.	Apply the MS Power point for practical applications	K3

SYLLABUS

1.	Introduction to Computer Hardware and Software Concepts
2.	MS Word : Document creation, Text manipulation with Scientific Notations
3.	MS Word :Table creation, Table formatting and Conversion
4.	MS Word: Advertisement Creation
5.	MS Word : Mail Merge Creation
6.	MS Word : Drawing Flowcharts
7.	MS Word : Creating CHARTS – Line, XY, Bar and Pie
8.	MS Excel: Create Employee work detail list
9.	MS Excel : Create charts
10.	MS Excel : Create Marks list
11.	MS Power point : Create Power Point presentation
12.	MS Power point : Create Advertisement

Text Books:

1.	Computer Fundamentals ñ Pradeep .K.Sinha: BPB Publications.
2.	Fundamentals of Computers by Reema Thareja from Oxford University Press
3.	Microsoft Office 2007 Fundamentals, 1st Edition By Laura Story, Dawna Walls

Reference Books:	
1.	Rajaraman, Introduction to Information Technology, PHI
2.	Introduction to Computers ñ Peter Norton Mcgraw Hill.
3.	Microsoft Excel 2007, Custom Guide Inc, 2007

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B20MC2102	MC	2	--	--	--	--	--	--

PROFESSIONAL ETHICS AND HUMAN VALUES

Common to CE, EEE, ME, AIDS & CSBS

Course Objectives: On completing this course student will be able to

1	Create awareness on Engineering Ethics and Human Values.
2	Instill Moral and Social Values and Loyalty.
3	Appreciate the rights of others.

Course Outcomes: By the end of the course, the student should have the ability to:

S.No	Outcome	Knowledge Level
1	Identify and analyze an ethical issue in the subject matter under investigation or in a relevant field.	K3
2	Identify the multiple ethical interests at stake in a real-world situation or practice.	K2
3	Articulate what makes a particular course of action ethically defensible.	K2
4	Assess their own ethical values and the social context of problems.	K3
5	Identify ethical concerns in research and intellectual contexts, including academic integrity, use and citation of sources, the objective presentation of data, and the treatment of human subjects.	K2
6	Demonstrate knowledge of ethical values in non-classroom activities, such as service learning, internships, and field work.	K3
7	Integrate, synthesize, and apply knowledge of ethical dilemmas and resolutions in academic settings, including focused and interdisciplinary research.	K4

SYLLABUS

UNIT-I (8 Hrs)	**Human Values:** Morals, Values and Ethics-Integrity-Work Ethic-Service learning Civic Virtue Respect for others Living Peacefully Caring Sharing Honesty -Courage-Cooperation Commitment Empathy Self Confidence Character Spirituality..
UNIT-II (8 Hrs)	**Engineering Ethics:** Senses of 'Engineering Ethics-Variety of moral issued- Types of inquiry Moral dilemmas Moral autonomy- Kohlberg's theory- Gilligan's theory-Consensus and controversy Models of professional roles-Theories about right action-Self-interest -Customs and religion Uses of Ethical theories Valuing time Cooperation Commitment.,
UNIT-III (8 Hrs)	**Engineering as Social Experimentation:** Engineering As Social Experimentation- Framing the problem- Determining the facts codes – of Ethics- Clarifying Concepts- Application issues Common Ground -General Principles-Utilitarian thinking respect for persons

UNIT-IV (8 Hrs)	**Engineers Responsibility for Safety and Risk:** – Safety and risk Assessment of safety and risk. Risk benefit analysis and reducing risk-Safety and the Engineer-Designing for the safety- Intellectual Property rights(IPR).,
UNIT-V (8 Hrs)	**Global Issues:** Globalization- Cross-culture issues-Environmental Ethics- Computer Ethics Computers as the instrument of Unethical behavior Computers as the object of Unethical acts Autonomous Computers-Computer codes of Ethics- Weapons Development -Ethics and Research Analyzing Ethical Problems in research.

Text Books:

1.	Engineering Ethics includes Human Values" by M.Govindarajan, S.Natarajan- and, V.S.Senthil Kumar-PHI Learning Pvt Ltd-2009.
2.	"Engineering Ethics" by Harris, Pritchard and Rabins, CENGAGE Learning, India Edition, 2009.
3.	"Ethics in Engineering" by Mike W. Martin and Roland Schinzinger - Tata McGraw-Hill-2003.
4.	"Professional Ethics and Morals" by Prof.A.R.Aryasri, DhanikotaSuyodhana-Maruthi Publications.
5.	"Professional Ethics and Human Values" by A.Alavudeen, R.Kalil Rahman and M.Jayakumaran-LaxmiPublications.
6.	"Professional Ethics and Human Values" by Prof.D.R.Kiran-
7.	"Indian Culture, Values and Professional Ethics" by PSR Murthy- BS Publication.
8.	Professional Ethics by R.Subramaniam - Oxford publications, New Delhi

Regulation: R20	II / IV - B.Tech. II - Semester								
CIVIL ENGINEERING									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2020-21 admitted Batch onwards)									
Course Code	**Course Name**	**Category**	**Cr**	**L**	**T**	**P**	**Int. Marks**	**Ext. Marks**	**Total Marks**
B20 BS 2204	Complex Variables and Statistical Methods	BS	3	3	0	0	30	70	100
B20 CE 2201	Highway Engineering	PC	3	3	0	0	30	70	100
B20 CE 2202	Structural analysis	PC	3	3	0	0	30	70	100
B20 CE 2203	Hydraulics and Hydraulic machinery	PC	3	3	0	0	30	70	100
B20 CE 2204	Design of Reinforced Concrete Structures	PC	3	3	0	0	30	70	100
B20 CE 2205	Geographic information system LAB	ES	1.5	0	0	3	15	35	50
B20 CE 2206	Environmental Engineering Lab	PC	1.5	0	0	3	15	35	50
B20 CE 2207	Fluid Mechanics and Hydraulic Machinery Lab	PC	1.5	0	0	3	15	35	50
B20 CE 2208	Advanced Surveying Lab	SOC	2	1	0	2	0	50	50
B20 MC 2201	English Proficiency	MC	0	2	0	0	--	--	--
		TOTAL	**21.5**	**18**	**0**	**11**	**195**	**505**	**700**

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B20BS2204	**BS**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

COMPLEX VARIABLES AND STATISTICAL METHODS

Common to CE & EEE

Pre-requisites: Basic concepts of Probability theory and Baye's Theorem

Course Objectives: Students are expected to

1	Learn the concept of Analytic function and its implications. Applications in Electrostatics and fluid flow problems.
2	Learn the concepts in complex integration and evaluation of real definite integrals.
3	Formulate and solve linear difference equations.
4	Learn important concepts of Z-transform and their use to solve linear difference equations.
5	Know the concepts of discrete and continuous random variables, learn a few important discrete/ continuous probability distributions
6	Learn Concepts of Sampling theory and develop a framework for testing of hypothesis for getting inferences about Population Parameters.

Course Out Comes: At the end of the course students will be able to

S. No	OUT COME	Knowledge Level
1	Comprehend the concept of Analytic function and apply in Electrostatics and Fluid dynamics	K3
2	Determine Laurent series of functions about isolated singularities, and determine residues. Use the residue theorem to evaluate certain real definite integrals.	K3
3	Formulate and solve linear difference equations.	K3
4	Use Z-transforms to solve linear difference equations with constant coefficients.	K3
5	Identify a random variable as discrete/continuous, find its expected value and also fit a probability distribution for a given frequency distribution.	K3
6	Decide the test applicable and apply it for giving inference about Population Parameter based on sample statistic for some large samples and small samples.	K3

SYLLABUS

UNIT-I (12 Hrs.)	**Functions of a Complex Variable** Limit and continuity of a function of a complex variable, derivative, analytic function, entire function, Cauchy- Riemann equations, determine an analytic function based on the knowledge of its real and imaginary parts, Milne-Thomson method, Applications of analytic function to flow problems, and in Electrostatics. Conformal mapping: the transformations defined by $w = z+c$, $w = cz$, $w = 1/z$. The Bilinear transformation.

UNIT-II (10 Hrs.)	**Complex Integration:** Line integral, Cauchy's integral theorem, Cauchy's integral formula. Expansion of a function in a Taylor series, McLaren series and Laurent series. Types of singularities, Residues, Cauchy's residue theorem. Evaluation of real definite integrals -integration around unit circle (Theorems without proofs).
UNIT-III (14 Hrs)	**Difference equations and Z-transforms:** Formation of a difference equation, Rules for finding complimentary function and particular integral for linear difference equations. Definition of Z- transform, some standard Z- transforms, properties, Shifting U_n to the right and to the left, transform of a function multiplied by n, initial value theorem and final value theorem(without proof), evaluation of inverse Z- transforms, convolution theorem (without proof), solution of linear difference equations by the use of Z-transforms.
UNIT-IV (10 Hrs.)	**Probability Distributions:** A brief review of random variables, Binomial, Poisson and Normal distributions, definitions of pmf/ pdf, notation, mean, variance, moment generating function. Fitting of Binomial or Poisson distributions for a given frequency distribution.
UNIT-V (12 Hrs.)	**Sampling theory and Testing of Hypothesis:** **Sampling theory:** Introduction, population and samples, level of significance, procedure o testing of hypothesis. **Large samples:** Testing of hypothesis for single proportion and two proportions. **Small samples:** Degrees of freedom, Students' t- distribution, t-test for single mean, two means; Chi- squared distribution, test for goodness of a fit.
TEXTBOOKS:	
1.	Scope and Treatment as in "Higher Engineering Mathematics", by Dr.B.S.Grewal, 43rd Edition, Khanna Publishers.
2.	Probability and statistics for Engineers, Miller and Freund, 7th edition, Pearson 2008.
REFERENCE BOOKS:	
1.	Fundamentals of Mathematical Statistics by S.C.Gupta and V.K.Kapoor, Sultan Chand & Sons Publishers
2.	Probability and statistics for Engineers and Scientists by Ronald E. Walpole, Sharon L. Myers and Keying Ye, 8th edition, Pearson Education, 2007.
3.	Advanced Engineering Mathematics, by Erwin Kreyszig,Wiley
4.	Higher Engineering Mathematics, by B.V.Ramana, Tata Mc Graw Hill Company.
5.	A text book of Engineering Mathematics, by N.P.Bali and Dr. Manish Goyal, Lakshmi Publications.
6.	Advanced Engineering Mathematics, by H.K.Dass, S.ChandCompany.
7.	Higher Engineering Mathematics, by Dr. M.K.Venkatraman, the National Publishing Company

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CE2201	PC	3	--	--	3	30	70	3 Hrs.

HIGHWAY ENGINEERING

(For CE)

Course Objectives: Students are expected to learn

1.	To impart different concepts in the field of Highway Engineering.
2.	To acquire design principles of Highway Geometrics and Pavements.
3.	To learn various traffic management plans.
4.	To acquire knowledge about pavement materials and design of flexible and rigid pavements.
5.	To learn various highway construction and maintenance procedures.

Course Outcomes At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	Plan the alignment of highway network for the given area.	K3
2.	Design the highway geometrical elements.	K3
3.	Design intersections and prepare traffic management plans.	K3
4.	Identify the suitability of pavement materials and design flexible & rigid pavements.	K3
5.	Understand the principles of construction and maintenance of highways.	K3

SYLLABUS

UNIT-I (8Hrs)	Highway Planning and Alignment: Highway development in India; Classification of Roads; Road Network Patterns; Necessity for Highway Planning; Different Road Development Plans – First, second, third road development plans, road development vision 2021, Rural Road Development Plan – Vision 2025; Planning Surveys; Highway Alignment- Factors affecting Alignment- Engineering Surveys – Drawings and Reports.
UNIT-II (8 Hrs)	Highway Geometric Design: Importance of Geometric Design- Design controls and Criteria Highway Cross Section Elements- Sight Distance Elements-Stopping sight Distance, Overtaking Sight Distance and Intermediate Sight Distance- Design of Horizontal Alignment-Design of Super elevation and Extra widening- Design of Transition Curves- Design of Vertical alignment Gradients- Vertical curves.
UNIT-III (10 Hrs)	Traffic Engineering: Basic Parameters of Traffic-Volume, Speed and Density- Traffic Volume Studies; Speed studies –spot speed and speed & delay studies; Parking Studies; Road Accidents - Causes and Preventive measures - Condition Diagram and Collision Diagrams; PCU Factors, Capacity of Highways – Factors Affecting; LOS Concepts; Road Traffic Signs; Road markings; Types of Intersections; At-Grade Intersections – Design of Plain, Flared, Rotary and Channelized Intersections; Design of Traffic Signals –Webster Method –IRC Method.
UNIT-IV (8 Hrs)	Highway Materials: Sub grade soil: classification –Group Index – Sub grade soil strength – California Bearing Ratio – Modulus of Sub grade Reaction. Stone aggregates: Desirable

	properties – Tests for Road Aggregates – Bituminous Materials: Types – Desirable properties – Tests on Bitumen – Bituminous paving mixes: Requirements – Marshall Method of Mix Design. Design Of Pavements: Types of pavements; Functions and requirements of different components of pavements; Design Factors Flexible Pavements: Design factors – Flexible Pavement Design Methods – CBR method – IRC method – Burmister method – Mechanistic method – IRC Method for Low volume Flexible pavements. Rigid Pavements: Design Considerations – wheel load stresses – Temperature stresses – Frictional stresses – Combination of stresses – Design of slabs – Design of Joints – IRC method – Rigid pavements for low volume roads – Continuously Reinforced Cement Concrete Pavements – Roller Compacted Concrete Pavements.
UNIT-V (6 Hrs)	Highway Construction and Maintenance: Types of Highway Construction – Earthwork; Construction of Earth Roads, Gravel Roads, Water Bound Macadam Roads, Bituminous Pavements and Construction of Cement Concrete Pavements. Pavement Failures, Maintenance of Highways, pavement evaluation, strengthening of existing pavements.

Textbooks:

1.	Highway Engineering, Khanna, S.K., Justo, C.E.G and Veeraragavan, A, Revised 10th Edition, Nem Chand & Bros, 2017.
2.	Traffic Engineering and Transportation Planning, Kadiyali L. R, Khanna Publishers, NewDelhi.

Reference Books:

1.	Principles of Highway Engineering, Kadiyali L. R, Khanna Publishers, NewDelhi
2.	Principles of Transportation Engineering, ParthaChakroborthy and Animesh Das, PHI Learning Private Limited, Delhi
3.	Highway Engineering, Paul H. Wright and Karen K Dixon, Wiley Student Edition, Wiley India (P) Ltd., New Delhi
4.	Transportation Engineering - An Introduction, JotinKhisty C, Prentice Hall, Englewood Cliffs, New Jersey.
5.	Traffic & Highway Engineering by Nicholas J. Garber, Lester A. Hoel, Fifth Edition, published in 2015,CENGAGELearning, New Delhi.
6.	Transportation Engineering and Planning, Papacostas C.S. and P.D. Prevedouros, Prentice Hall of India Pvt.Ltd; New Delhi.
7.	Highway Engineering, Srinivasa Kumar R, Universities Press, Hyderabad
8.	Practice and Design of Highway Engineering, Sharma S. K., Principles, S. Chand & Company Private Limited, New Delhi.
9.	Highway and Traffic Engineering, Subhash C. Saxena, CBS Publishers, New Delhi.
10.	Transportation Engineering Volume I by C. Venkatramaiah, Universities Press, New Delhi.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CE2202	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

STRUCTURAL ANALYSIS

(For CE)

Course Objectives: Students are expected to learn

1.	To familiarize with the deflection of simple determinate beams frames & trusses
2.	To impart knowledge on method of consistent deformation and energy methods.
3.	To analyze the Indeterminate continuous beams and Rigid frames by force and displacement methods.
4.	To draw the influence lines and moving loads for statically determinate beams.

Course Outcomes At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	Determine deflections indeterminate beams ,frames & trusses by different methods and apply strain energy concept	K3
2.	Analyze statically indeterminate beams by method of consistent deformation	K3
3.	Analyze Statically indeterminate continuous beams by theorem of three moments and rigid frames by force method	K3
4.	Analyze Statically indeterminate continuous beams and rigid frames by slope deflection method.	K3
5.	Determine reactions, BM&SF in beams subjected to moving loads using ILD	K3

SYLLABUS

UNIT-I (12Hrs)	**Deformation of statically determinate beams** by using (i) Moment area method(ii)Conjugate beam method(iii)Unit load method Strain- energy due to (i) Axial load (ii) Bending Moment (iii) Shear force and (iv) Torque **Deformation of Statically Determinate rigid frame and trusses:** (i) Determination of slopes and deflection of statically ***determinate rigid frames*** using Unit load method Determination of deflection of statically ***determinate trusses*** using Unit load method
UNIT-II (8 Hrs)	**Analysis of Statically indeterminate beam** a) Analysis of propped cantilever beams and fixed beams subjected to concentrated load and uniform distributed load by method of consistent deformation. Analysis of two span continuous beams by using law of Reciprocal Deflection and theorem of least work and induced reaction due to yielding of supports.
UNIT-III (12 Hrs)	**Analysis of Statically indeterminate continuous beam and rigid frames by force method.** a) Analysis of three span continuous beams **due to applied loads** by theorem of three moments b) Analysis of three span continuous beams **due to uneven support settlements** by

<table>
<tr><td></td><td>theorem of three moments
c) Analysis of statically indeterminate rigid frames due to applied loads by force method.
Analysis of induced reactions on statically indeterminate rigid frames due to yielding of supports by force method.</td></tr>
<tr><td colspan="2"></td></tr>
<tr><td>UNIT-IV
(10 Hrs)</td><td>Analysis of Statically indeterminate continuous beam and rigid frames
a) Analysis of statically indeterminate three span continuous beams due to applied load by slope deflection method.
b) Analysis of statically indeterminate three span continuous beams for uneven support settlements by slope deflection method.
c) Analysis of statically indeterminate rigid frames (without side sway and with side sway) by slope deflection method.
Analysis of statically indeterminate rigid frames due to yielding of supports by slope deflection method.</td></tr>
<tr><td colspan="2"></td></tr>
<tr><td>UNIT-V
(8 Hrs)</td><td>Influence Lines
Definition – Influence line for Reaction, SF and BM in case of simply supported and overhanging beams-Load position for Max SF at a section –Load position for max BM at a section- Single point load, U.D.L longer than the span, U.D.L. shorter than the span.
Moving loads – Max SF and BM at a given section and absolute Max SF and BM due to single concentrated load, U.D.L. longer than the span, U.D.L. shorter than the span ,two point loads with fixed distance between demand several point loads, Maximum Bending moment at a section under a wheel load and absolute maximum Bending moment in the case of several wheel loads-Equivalent uniformly Distributed load, Focal length.</td></tr>
</table>

Textbooks:	
1.	Basic Structural Analysis, C.S. Reddy, Mc Graw Hill Education(India) PVT.LTD
2.	Structural Analysis- T.S. Thandavamoorty, Oxford University Press, New Delhi
Reference Books:	
1.	Statically indeterminate structures – C.K. Wang, Mc Graw Hill Education PVT.LTD
2.	Mechanics of structures Vol. II- S.B.Junnarkar and Dr.H.J.shah, Charotar Publishing House.
3.	Structural analysis – Devdas Menon, Narosa Publishing House PVT.LTD
4.	Theory of Structures Volume II, S.P Gupta, G.S Pandit, R. Gupta – Tata McGraw-Hill Publishing Company Limited, New Delhi

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CE2203	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

HYDRAULICS AND HYDRAULIC MACHINERY

(For CE)

Course Objectives: Students are expected to learn

1.	Understand the fundamental concept of dimensional analysis, Model Laws and its applications
2.	Understand the concept of Boundaries layers thickness and Determine drag force for various bodies, Lift force on circular cylinder and Air foil
3.	Understand the design philosophy and Characteristic curves of turbines and pumps
4.	Understand the Uniform and Non–Uniform flow in Open Channel

Course Outcomes At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	Apply the principles of dimensional homogeneity and Similarity laws for irrigation structures and fluid Machinery.	K3
2.	Determine the Drag and Lift force for fully submerged bodies.	K3
3.	Use momentum and energy principles for design of turbines	K3
4.	Use momentum and energy principles for design of pumps	K3
5.	Determine the discharge of most economical channel section for uniform flow in open Channels and Specific Energy, Critical flow, critical depth and critical velocity.	K3

SYLLABUS

UNIT-I (10Hrs)	**Dimensional Analysis and Similitude:** Dimensional Homogeneity-Methods of Dimensional Analysis – Rayleigh's Method – Buckingham's π theorem – Superfluous and Omitted Variables - Similitude – Model Analysis – Dimensionless numbers – Similarity Laws– Model testing of partially submerged bodies– Types of models. Boundary Layer Theory: Introduction–characteristics of laminar boundary layer–boundary layer growth over a flat plate (without pressure gradient)–Boundary thicknesses – Stability parameter – Turbulent boundary layer – boundary layer separation – Boundary layer on rough surfaces–laminar sub layer.
UNIT-II (10 Hrs)	Flow past submerged bodies: Introduction–Types of Drag–Drag on a sphere–Drag on a cylinder – Von Karman Vortex Trail – Drag on a flat plate – Development of Lift on immersed circular cylinder, air foil– Magnus effect. Impact of Jets: Impulse momentum equation – Momentum Correction factor, Force on Stationary flat plate – moving flat plate - Force on Stationary curved vanes – moving curved vanes.
UNIT-III (10 Hrs)	**Hydraulic Turbines**: Introduction - Classification based on Head, Discharge, Hydraulic Action–Impulse and Reaction Turbines, Differences between Impulse and Reaction Turbine, Choice of Type of Turbine, Component Parts & Working principle of a Pelton

	Turbine, Francis Turbine-Velocity Triangles-Hydraulic and Overall efficiencies. Performance of turbines: Performance under Unit head, power and speed – Performance under specific conditions - Specific Speed and its importance. Performance Characteristic Curves– Operating Characteristic Curves– Cavitation -Draft Tube.
UNIT-IV (10 Hrs)	**Centrifugal Pumps**: Types of Pumps–Selection Criterion–Comparison between Centrifugal & Reciprocating Pumps-Centrifugal Pumps–Component Parts & Working Principle–Classification of Centrifugal pumps-Cavitation–Maximum Suction lift–NPSH. Specific Speed of pumps–Performance Characteristics of Centrifugal Pumps–Dimensionless characteristics – Constant efficiency curves of Centrifugal Pumps Reciprocating Pumps: Component Parts–Working Principle of single acting and double acting reciprocating pumps–Discharge Co-efficient, Volumetric efficiency and Slip. Work done and Power Input–Indicator Diagram, Effect of acceleration and friction on Indicator Diagram-Air Vessels.
UNIT-V (8 Hrs)	**Flow through Open Channels:** Classification of open channels, Uniform Flow: Chezy's and Manning's formula, Hydraulic mean depth, hydraulic radius. Most economical trapezoidal and rectangular channel section – Specific energy, Critical Flow. G.V.F: length of surface profile using step by step Method, M1, M_2, M_3, S1, S2 and S3 CURVES. Steady Rapidly Varied Flow: Hydraulic Jump in a horizontal rectangular channel, Specific force Computation of energy loss.

Textbooks:

1.	Modi,P.N. &Seth,S.M.(2009),"Fluid Mechanics and Hydraulic Machinery ",StandardBook House,New Delhi,19thEdition.
2.	Jain,A.K.(2008),"Fluid Mechanics",KhannaPublishers,NewDelhi,4thEdition

Reference Books:

1.	Kumar,K.L.,Chand,S.&Co.(2008),"EngineeringFluidMechanics",EurasiaPublishingHouse(P) Ltd,NewDelhi,8th Edition.
2.	Subramanyam,K.(2008),"FlowinOpenChannels",McGrawHillEducation,NewDelhi,3rd Edition.
3.	Chow,V.T.(2009),"Open-ChannelHydraulics",TheBlackburnPress,Caldwell,NJUSA,1st Edition
4.	White,F.M.(2011)"FluidMechanics",TataMcGrawHillPublication,NewDelhi,7th Edition
5.	RelevantNPTELCourses

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CE2204	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

DESIGN OF REINFORCED CONCRETE STRUCTURES

(For CE)

Course Objectives: Students are expected to learn

1.	To introduce the different types of philosophies related to design of basic structural elements such as slab, beam, column and footing which form part of any structural system with reference to Indian standard code of practice.

Course Outcomes At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	Understand the various design methodologies for the design of RC elements. Analyze and design the flexural members.	K3
2.	Design the reinforced concrete beams subjected to shear only and also combined action of shear and torsion.	K3
3.	Distinguish between the behavior of one way and two way actions in slab and familiarize to design of two way slabs whose corners restrained and not restrained from lifting up	K3
4.	Design compression members.	K3
5.	Design stair case and footing.	K3

SYLLABUS

UNIT-I (10Hrs)	Introduction: Loading standards as per IS 875, Grades of steel and cement, Introduction to basic design concepts like Working Stress Method (W.S.M), Ultimate Load Method (U.L.M) and Limit State Method (L.S.M.). Limit State of Collapse: Introduction, Characteristic load and strengths, Design values, PartialsafetyfactorsLoadsandmaterials,Stress-straincharacteristics of concrete and steel. Limit State of Collapse in Flexure: Assumptions, Limiting depth of neutral axis. Concrete stress block in compression. Under reinforced, Balanced and over reinforced sections. Analysis of singly Reinforced rectangular sections, analysis of singly reinforced flanged section, analysis of doubly reinforced rectangular sections. Code requirement for design of flexural reinforcement are effective span, concrete cover, spacing of reinforcing bars, minimum and maximum areas of flexural reinforcement, requirements for deflection control, general guide lines for choosing beam size. Design of singly and doubly reinforced rectangular sections. Estimation of Effective flange width, Design of flanged beams (T-Beams).
UNIT-II (10 Hrs)	Limit State of Collapse in Shear**:** Shear stresses in homogeneous rectangular beam, modes of cracking, shear transfer mechanisms, shear span/depth ratio, and shear failure modes. Calculation of nominal shear stress, critical sections for shear design, Design shear strength of concrete in beams with and without shear reinforcement. Types of shear reinforcement, Factors contributing to ultimate shear resistance, limiting ultimate shear resistance. Shear resistance of web reinforcement (Truss analogy), minimum shear reinforcement. Design of shear reinforcement in beams as perISCode456. Enhanced shear near supports. Steel detailing.

	Limit State of Collapse in Torsion: Need for torsion reinforcement, Reinforcement for torsion in RC beams, design strength in torsion combined with flexure and IS code provisions for design of longitudinal reinforcement, design strength in torsion combined with shear and IS code provisions for design of transverse Reinforcement, distribution of Torsion reinforcement. Design of rectangular section for combined bending shear and torsion. Detailing of torsion reinforcement. Limit State of Collapse in Bond**:** Concept of bond, Code requirement for bond, flexural bond, anchorage bond, development length. Bends, Hooks and Mechanical anchorages. Splicing of reinforcement, Lap Splices, Welded Splices and Mechanical Connections.
UNIT-III (10 Hrs)	**Design of one-way slabs:** Behavior of one-way slabs, general considerations for slabs, minimum flexural reinforcement in slabs, deflection control by limiting Span/Depth ratio. Design of simply supported one-way slabs and design of continuous one way slabs using moment coefficient, detailing of reinforcement in one-way slabs. **Design of two-way slab:** Behavior of two-way slabs, design of wall supported two-way slabs, slab thickness based on deflection control criterion, uniformly loaded and simply supported rectangular slabs (Rankine-Grash off theory), Uniformly loaded restrained rectangular slabs usingIS456provisions.Detailingofflexuralreinforcementandtorsionalreinforcement.Shear force in uniformly loaded two-way slabs.
UNIT-IV (10 Hrs)	**Limit State of Collapse in Compression:** Classification of columns based on type of reinforcement, type of loading and slenderness ratios. Estimation of effective length of a column-definition of effective length- unsupported length. Effective length ratios for idealized boundaryconditions- Coderecommendationsforidealizedboundaryconditions. Code requirements on slenderness limits, minimum eccentricities and reinforcement. Design of short column under axial compression-condition of axial loading- behavior under ultimate loads-Tied columns-Spiral columns. Design strength of axially loaded short columns. Design of Short columns subjected to combined axial load and uniaxial moments by using design hand book SP:16. Design of Short columns subjected to combined axial load and biaxial moments by using design handbook SP:16
UNIT-V (10 Hrs)	**Design of Staircases:** Types of staircases Geometrical Configurations, Structural Classification, stair slab spanning transversely, stair slab spanning longitudinally Loads and Load effects on stair Slabs, Design of stair slab spanning longitudinally. **Design of Footings:** Types of footing-Isolated footings, Soil pressures under isolated footings-Allowable soil pressure, Distribution of base pressure- Concentrically loaded footings, Instabilityproblems- Overturningandsliding.GeneraldesignconsiderationsandCoderequirements-Factored soil pressure at ultimate limit state, general design considerations, Thickness of footing base slab, Design for Shear, design for Flexure, Transfer of force at Column Base. Design of isolated square and Rectangular footings concentrically loaded.

Textbooks:	
1.	Reinforced concrete design by S.Unnikrishna Pillai&Devdas Menon,Tata Mc. GrawHill,New Delhi.
2.	Limit State Design by B.C.Punmia, Ashok Kumar Jain and Arun Kumar Jain, Laxmi publications Pvt. Ltd., New Delhi.
3.	Reinforced concrete Limit state design by Ashok K. Jain, NemChand &Bros, Roorkee.
4.	Limit state designed of reinforced concrete –P.C.Varghese,PrinticeHallofIndia,NewDelhi.
5.	Reinforced Concrete Vol.I(ElementaryReinforced concrete) by Dr.H.J.Shah,Charotar publication house Pvt.Ltd, ANAND,Gujarath.
Reference Books:	
1.	Fundamentals of Reinforced concrete design by M.L.Gambhir, Printice Hall of India Private Ltd. ,New Delhi.
2.	Reinforced concrete structural elements–behavior, Analysis and design by P.Purushotham, TataMc.Graw-Hill, 1994.
3.	Design of concrete structures–Arthus H.Nilson,David Darwin,andChorlesW.Dolar,TataMc.Graw-Hill,3rdEdition, 2005.
4.	Reinforcedconcretestructures,Vol.1,byB.C.Punmia,AshokKumarJainandArunKumarJain,Laxmi, publications Pvt.Ltd., New Delhi
5.	Reinforced concrete structures– I.C.Syal&A.K.Goel,S.ChandPublishers
6.	Reinforced concretedesignbyN.KrishnaRajuandR.N.Pranesh,NewageInternationalPublishers,New Delhi
7.	Design of Reinforced Concrete Structures by N. Subramanian, Oxford University Press, YMCA LibraryBuilding,1JailSingh Road, NewDelhi, India.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CE2205	**ES**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3 Hrs.**

GEOGRAPHIC INFORMATION SYSTEMS LABORATORY

(For CE)

Course Objectives: Students are expected to learn

1.	Familiarize the students with digital map creation life cycle
2.	Introduce and make students practice various satellite image processing techniques
3.	Understand the process of digitization, creation of thematic map
4.	Introduce advanced techniques like 3D modeling.

Course Outcomes: At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	Choose the datum and projection systems that suits the data	K3
2.	Handle the raw data	K3
3.	Create thematic layer by using on-screen digitization techniques and attaching attribute data	K3
4.	Visualize and Interpret digital elevation model	K3

EXCERCISES IN GIS

Fundamentals of Computers

1.	Digitization of Map / Toposheet	**[CO2]**
2.	Creation of thematic maps	**[CO3]**
3.	Estimation of features and their interpretation	**[CO3]**
4.	Developing Digital Elevation Model	**[CO4]**

SOFTWARES USED:

1.	QGIS
2.	Arc GIS 9.3 / 10.1
3.	ERDAS 9.3
	Any one or Equivalent.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CE2206	**PC**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3 Hrs.**

ENVIRONMENTAL ENGINEERING LAB

(For CE)

Course Objectives: Student shall be able to

1	To test physical properties of water
2	To test chemical properties of water
3	To know about the water quality testing as a whole

Course Outcomes: After the completion of the course student should be able to

S.No	Outcome	Knowledge Level
1	Determine physical properties of water	K5
2	Determine hardness, acidity and alkalinity of water	K5
3	Estimate chloride, available chlorine, BOD and COD	K5
4	Estimate solids present in water sample	K5

LIST OF EXPERIMENTS

1	Determine pH of water
2	Determine Hardness of water
3	Determine Acidity of water
4	Determine Alkalinity of water
5	Determine Chlorides in water sample
6	Determine DO in water sample
7	Determine BOD in water sample
8	Determine Estimation of total solids, suspended solids, dissolved solids in water sample
9	Jar test
10	Electrical conductivity

Reference Books:

1	Environmental Engineering by S.K.Garg

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CE2207	**PC**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3 Hrs.**

FLUID MECHANICS AND HYDRAULIC MACHINERY LAB

(For CE)

Course Objectives : Student shall be able to

1.	verify the principles of channel flow in laboratory by conducting experiments
2.	Aware of tests to identify the different types of flow measuring devices to measure flow in tanks, Pipes and open channels
3.	Aware of various types of pumps and turbines

Course Outcomes: After the completion of the course student should be able to

S.No	Outcome	Knowledge Level
1.	Illustrate Flow Measuring Devices used in pipes, channels and Tanks	K3
2.	Analyze characteristics of broad crested notch	
3.	Determine the coefficient of impact on a flat plate and curved vane by comparing The theoretical and actual forces by impact.	K3
4.	Analyze the working of the reciprocating pump and centrifugal pump and develop the characteristics of power input, head and efficiency under various discharges And plot the characteristic curves.	K4
5.	Determine the performance characteristics of pelton wheel turbine and develop the characteristic curves of unit discharge, unit power and unit head under varying Unit speed	K3
6.	Determine the performance characteristics of Francis turbine and develop the characteristic curves of unit discharge, unit power and unit head under varying Unit speed	K3

LISTOFEXPERIMENTS

1.	Calibration of Venturimeter & Orifice meter
2.	Determination of Coefficient of discharge for a small orifice and mouth piece by a constant head and variable head method.
3.	Calibration of contracted Rectangular Notch and/or Triangular Notch
4.	Determination of Coefficient of loss of head in a sudden contraction and friction factor
5.	Verification of Bernoulli's equation
6.	Impact of jet on vanes
7.	Study of Hydraulic jump
8.	Efficiency test on centrifugal pump
9.	Efficiency test on reciprocating pump
10.	Performance test on Pelton wheel turbine
11.	Performance test on Francis turbine

Reference books:

1.	Hydraulic Fluid Mechanics and Fluid Machines,S.Ramamrutham,DhanpatRaiPublishingCo.
2.	Fluid Mechanics and Hydraulic Machines byR.K.Bansal, LaxmiPublications

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CE2208	**SOC**	**1**	**--**	**2**	**2**	**--**	**50**	**3 Hrs.**

ADVANCED SURVEYING LAB

(For CE)

Course Objectives: Student shall be able to

1.	Prepare the student to plan and conduct field work and application of scientific Methodology in handling field samples.
2.	Interpret the False Color Composite Images of Satellite data.
3.	Create Fundamental Knowledge on GIS software.
4.	Learn the identification of Geographic Coordinates by GPS.
5.	Study the land use/ Land cover dynamics of certain region.

Course Outcomes: At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	Fully equipped with various surveying concepts and methods using advanced ground survey equipment's.	K3
2.	Carry out profiling and grid levelling, for generation of profiles, contour maps, and earth works computations.	K3
3.	Handle the Satellite images and interpret the satellite data.	K3
4.	The interpret data can be used to prepare plan for urban development/town planning.	K3
5.	Prepare the candidates with National Global employability.	K3

LIST OF EXPERIMENTS

1.	Transferring and Drafting the collected raw data from total station survey using AutoCAD.
2.	Computation on drafted data using AutoCAD.
3.	Developing Contour using raw data from total station using Surfer software.
4.	Visual Interpretation of standard FCC (False Color composite).
5.	Digitization of physical features on a map/image using GIS software.
6.	Coordinates measurement using GPS.

Reference books:

1.	AMChandra:Higher surveying
2.	TM Lillesand etal:Remote sensing&Imag eInterpretation
3.	B.Bhatta:Remote sensing &GIS
4.	M.Anjireddy:Remote sensing& GIS,BS Publications
5.	NKAgarwal:essentials of GPS, SpatialNetworks,Hyderabad

Code	Category	L	T	P	C	I.M	E.M	Exam
B20MC2201	**MC**	**2**	**--**	**--**	**--**	**--**	**--**	**--**

ENGLISH PROFICIENCY

(Common to CE,EEE,ME,AIDS & CSBS)

Course Objectives: The students will be able to

1.	Communicate their ideas and views effectively
2.	Practice language skills and improve their language competency.
3.	Know and perform well in real life contexts
4.	Identify and examine their self-attributes which require improvement and motivation.
5.	Build confidence and overcome their inhibitions, stage freight, nervousness etc.,
6.	Improve their reading skills.

Course Outcomes: At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	Improve speaking skills.	K3
2.	Enhance their listening capabilities	K3
3.	Learn and practice the skills of composition writing.	K3
4.	Enhance their reading and understanding of different texts.	K3
5.	Improve their communication both in formal and informal contexts.	K3
6.	Be confident in presentation skills.	K3

SYLLABUS

UNIT-I	**Listening Skills** Types of listening Hearing and Listening Listening as a receptive skill
UNIT-II	**Speaking Skills** Presentation skills Describing event/place/thing Extempore Debate Group Discussion
UNIT-III	**Reading Skills** Types of Reading (Intensive and Extensive reading, Skimming, Scanning) Reading/Summarizing News Paper Articles
UNIT-IV	**Writing Skills** Essay Writing (Argumentative, Analytical and Descriptive) E-Mail Writing Business Letters Resume Writing

UNIT-V	**Integrated Language Skills** Listening Skills for Speaking and Writing Reading Skills for Writing and Speaking
Reference Books:	
1.	Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP
2.	English and Communication Skills for Students of Science and Engineering, by S.P. Dhanavel, Orient Blackswan Ltd. 2009
3.	Enriching Speaking and Writing Skills, Orient Blackswan Publishers.
4.	The Oxford Guide to Writing and Speaking by John Seely OUP.
5.	Effective Technical Communication by M.AshrafRizwi. Tata Mcgraw hill.
6.	Six Weeks to Words of Power by Wilfred Funk. W.R.Goyal Publishers
Note: Internal Assessment is carried out throughout the semester.	